

	B_0 (0..31)	B_1 (32..63)	B_2 (64..95)
T_1	$C_1T_1B_0$ (SQ8 [32B])	$C_1T_1B_1$ (SQ8 [32B])	$C_1T_1B_2$ (SQ8 [32B])
T_2	$C_1T_2B_0$ (SQ8 [32B])	$C_1T_2B_1$ (SQ8 [32B])	$C_1T_2B_2$ (SQ8 [32B])

Regime ($W_0 = 9.0 \times 10^8$)	Selected Kernel Path
$W < W_0$ ($B \le 32$, NUS)	Single-Pass CM-FP32 (1.00 ×)
$W \ge W_0$ (High-Work)	3-Stage PCA+SQ8 (14.6 ×)

(a) Cluster-major blocked layout & online router

Step (1) Warm-up & probed cells

Query	Warm $m_{\text{warm}} = 2$	Threshold $\tau_K^{(0)}$	Target Cell
Q_1	C_1, C_2 (FP32)	$\tau_K = 0.420$	C_3 ($x_1..x_4$)

Per-cluster SQ8 intervals (C_3)

Block m	PCA Dims	Bounds $[\ell_{c,d}, \delta_{c,d}]$
$m = 0, 1$	0..63	1D LB $\psi_{c,d}(\tilde{q}_d, \hat{x}_d)$

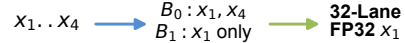
Step (2) Progressive 32-dim SQ8 lower-bound cascade ($L_b(\tilde{q}, \hat{x})$ vs. $\tau_K = 0.420$)

Candidate	x_1 (C_3)	x_2 (C_3)	x_3 (C_3)	x_4 (C_3)
Block $m = 0$ (0..31)	$L_1 = 0.130 \le 0.420$ [Survives]	$L_1 = 0.456 > 0.420$ [Pruned at $m = 0$!]	$L_1 = 0.507 > 0.420$ [Pruned at $m = 0$!]	$L_1 = 0.298 \le 0.420$ [Survives]
Block $m = 1$ (32..63)	$L_2 = 0.192 \le 0.420$ [Survives to B_{blk}]	Skipped (96.7% dims)	Skipped (96.7% dims)	$L_2 = 0.448 > 0.420$ [Pruned at $m = 1$!]

Step (3) Stage Z Warp-cooperative exact FP32 refinement

Survivor	All-Block LB $L_{B_{\text{blk}}}$	32-Lane Exact FP32 $\|q_1 - x_1\|_2^2$	Heap Update
x_1 only	$0.371 \le 0.420$	$0.384 < 0.420$ (100.0% Exact)	$\tau_K \leftarrow 0.384$

Warp lane flow



(b) Progressive 32-dim block pruning trace ($\tau_K = 0.420$)